

Dr. Courtney B. Watson

Astronomer | Project & Data Management Consultant

CONTACT

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🌐 Courtney Watson
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📡 NASA/ADS publication list

EDUCATION

Ph.D., Astrophysics 📅 2022-2025
📍 Boston University

M.A., Astrophysics 📅 2019-2022
📍 Boston University

B.S., Physics 📅 2015-2018
📍 Texas A&M University

SKILLS

Programming

Python ●●●●●●●●
HTML/CSS ●●●●●●●●
SQL ●●●●●●●●
Java ●●●●●●●●
LaTeX ●●●●●●●●

Software & Tools

Data handling/analysis (e.g. numpy, scipy, pandas...) ●●●●●●●●

Visualization (e.g. matplotlib, bokeh, seaborn...) ●●●●●●●●

Project Management (e.g. Airtable, Smartsheet, Project...) ●●●●●●●●

Version Control (e.g. git) ●●●●●●●●

AREAS OF EXPERTISE

Project & Data Management

Technical and Grant Writing

Cross-functional Collaboration

Data Visualization

Technical Analysis

ABOUT ME

Detail-oriented, PhD-trained scientist with a strong foundation in data analysis and project management, complemented by advanced skills in statistical analysis and visualization. With over seven years of experience, I have honed my ability to communicate complex findings clearly to diverse audiences. My background includes managing data-driven projects, developing research methodologies, and mentoring students, which aligns well with roles requiring strategic advisory and technical expertise. I am proficient in Python, SQL, and data visualization tools, and have a proven track record of collaborating with cross-functional teams to achieve project milestones. Eager to transition to industry, I am looking to leverage my analytical and problem-solving skills in data science or analytics roles, where I can contribute to innovative solutions and drive impactful results.

WORK HISTORY

📅 2026–now
📍 Harvard & Smithsonian CfA
Project & Data Management Consultant
Continuing contracted engagement supporting the AGEL survey, providing expert project and data management consulting to ensure continuity of research operations and strategic data workflows for international collaborators.

- Advising on data architecture and maintaining the permanent archive with established access and upload protocols.
- Delivering project coordination support, tracking milestones and reporting progress to distributed research teams.
- Producing visual analytics and documentation materials to sustain data reproducibility and outreach efforts.
- Providing mentorship continuity for graduate and undergraduate researchers as needed.

📅 07/2025–03/2026
📍 Harvard & Smithsonian CfA
Postdoctoral Researcher
As Project and Data Manager for AGEL survey I developed and implemented strategic data management plans, ensuring efficient data handling and organization for research teams, while coordinating project milestones and reporting progress to international collaborators.

- Created a comprehensive team roadmap, detailing resources, data products, and onboarding procedures.
- Established a permanent data archive with defined access and upload protocols.
- Prepared visual analytics materials to enhance data reproducibility and outreach.
- Guided Master's and undergraduate students in their research, fostering skill development.

📅 2020–2025
📍 Boston University
Graduate Research Assistant
Conducted PhD research on galaxy clusters, focusing on thermodynamic properties and the role of active galactic nuclei within their environments. Utilized space-based data to develop methodologies for precise analysis, contributing to the understanding of complex astrophysical phenomena.

- Created software to process and analyze scientific data, enhancing research accuracy.
- Led projects from concept to completion, ensuring high-quality outcomes.
- Regularly presented findings in academic and public forums, gaining valuable feedback.
- Published research in peer-reviewed journals, advancing field knowledge.

📅 2020–2024
📍 Boston University
Graduate Teaching Assistant
As a Teaching Fellow, I designed and conducted lab sessions and discussions, enhancing students' understanding of complex astrophysics concepts. I also developed a comprehensive TA Knowledge Database to streamline grading processes and support fellow TAs.

- Led lab sessions and discussions to facilitate student learning in astrophysics.
- Created and managed a TA Knowledge Database, improving efficiency and reducing prep time.
- Awarded Outstanding Teaching Fellow for contributions to teaching and organizational improvements.